

P. KUMARESHAN

Precision Machining Specialist • CNC | VMC | HMC | 5-Axis Operations

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PROFESSIONAL SUMMARY

Results-driven CNC/VMC/HMC Machining Specialist with 6+ years of progressive experience in high-precision manufacturing environments. Demonstrated expertise in 5-axis VMC and HMC machine operation, setup, and process optimization at Tier-1 engineering manufacturers including Lakshmi Machine Works Limited and Texmo Industries. Proven ability to execute tight-tolerance machining operations, minimize cycle times, and sustain near-zero rejection rates. Proficient in CREO-based CNC/VMC programming, G-code operations, and precision metrology.

TECHNICAL SKILLS

Machine Operation: 5-Axis VMC, HMC, CNC Turning, CNC Grinding, Multi-Axis Machining

Programming: CNC Programming, VMC Programming, G-Code, M-Code, Tool Path Optimization

CAD / CAM Software: CREO (PTC), Part Design, Tool Path Generation, CAM Simulation

Setup & Tooling: Machine Setup, Tool Presetting, Fixture Configuration, Work Offset Setting

Quality & Inspection: Dimensional Inspection, Vernier Calipers, Micrometers, Bore Gauges, GD&T

Manufacturing: Lean Manufacturing, Preventive Maintenance, Process Improvement, First Article Inspection

PROFESSIONAL EXPERIENCE

5-Axis VMC & HMC Machine Operator / Setter | *Lakshmi Machine Works Limited (LMW)* | Coimbatore, Tamil Nadu | Dec 2024 – Present

- Operate and set up 5-axis VMC and HMC machines for high-precision aerospace and industrial component machining, consistently achieving dimensional tolerances within ± 0.01 mm.
- Execute end-to-end machine setup including fixture mounting, tool presetting, work offset configuration, and CNC program verification prior to each production run.
- Implement process optimizations on feed rates, depth of cut, and cutting parameters, contributing to improved machine utilization and measurable scrap reduction across production cells.
- Perform first-article inspections in coordination with quality engineers, ensuring full conformance to engineering drawings, GD&T specifications, and customer tolerances.
- Maintain detailed machine logs, production records, and shift handover reports to support lean manufacturing and continuous improvement (Kaizen) initiatives.

CNC Turning, Grinding & VMC/HMC Operator | *Texmo Industries* | Coimbatore, Tamil Nadu | Apr 2019 – Dec 2024 (5 Years 9 Months)

- Operated multi-axis CNC turning and grinding machines to manufacture pump components and precision parts, sustaining high-volume production throughput across 8-hour shift cycles.
- Programmed and validated CNC turning cycles and grinding parameters, optimizing feed rates and tool paths to reduce average cycle times and boost throughput efficiency.
- Executed VMC and HMC operations for complex multi-feature geometries, consistently maintaining quality output across batch runs of 50–500+ components.
- Conducted rigorous in-process dimensional inspections using precision measuring instruments (vernier calipers, micrometers, bore gauges), achieving near-zero rejection rates on critical tolerances.
- Contributed to preventive maintenance activities and new component trials, ensuring smooth production launches and on-time delivery to internal and external customers.

EDUCATION

Diploma in Mechanical Engineering (DME) — 82% Aggregate

Ramakrishna Mission Polytechnic College • DOTE, Tamil Nadu • 2018

SSLC – Secondary School Leaving Certificate — 80% Aggregate

Govt. Boys Hr. Sec. School, Pillipalaym • State Board • 2015

CERTIFICATIONS & TRAINING

- Skill Enhancement Program in Manufacturing – Industry-recognized upskilling program focused on precision machining and production engineering practices.
- 2-Year Personality Development Camp under SGVEP – Structured program covering professional conduct, teamwork, and workplace leadership.
- 2-Year Ramakrishna English Connect Program – Professional communication and technical English for engineering and manufacturing environments.

KEY STRENGTHS

- 6+ years of continuous hands-on experience across CNC, VMC, HMC, and multi-axis machining in Tier-1 manufacturing companies.
- Dual exposure to high-volume pump component production (Texmo Industries) and precision-critical textile machinery components (LMW).
- Skilled in reading and interpreting 2D engineering drawings, GD&T annotations, and part specifications to execute accurate machining setups.
- Consistent academic achiever (80–82%) with a strong foundation in Mechanical Engineering principles supporting effective shop-floor problem solving.

LANGUAGES

Tamil (Native) • English (Professional Working Proficiency)